

Notation

Formal languages, and the three that selfie is made of: EBNF, C*, and RISC-U.

← ICS 03 · Everything is Bits

The score is not the music.

SYNTAX · NOTATION

Marks on paper. Finite, discrete, checkable, copyable. 1 + 1, a staff of crotchets, H₂O, a line of code, this sentence.

SEMANTICS · MEANING

What the marks are *about*. The number two. Sound in a room. A molecule that dissolves salt. A machine's behaviour over all inputs.

A semantics is a function from notation to meaning. Defining that function precisely is the founding act of every exact discipline — and computer science is where it was done most sharply, because a machine does exactly what the marks say and nothing else.

Frege 1892 · Sinn / Bedeutung

Saussure · signifier / signified

Korzybski 1931 · the map is not the territory

Magritte 1929 · ceci n'est pas une pipe

TWO KINDS OF LANGUAGE

Natural language is **compression**.
Formal language is **commitment**.

"I saw the man with the telescope." Two readings, one string. English resolves it with a shared world you and I already have. That shared world is why English is powerful — and why it cannot be a foundation. Its semantics is *us*, and we differ.

A formal language pays a price — narrowness, pedantry, effort — to buy one thing: a meaning that does not depend on who is reading.

Prompting an AI in English is negotiation. Writing a test, a type, a schema, a unit, a contract is legislation.

NATURAL	FORMAL
ambiguous	single-valued
elastic, forgiving	brittle, exact
persuades	proves
needs a mind	needs a machine
learned by living	defined by decree

EBNF says what code may **look like**.
It can even say what EBNF looks like.

```
integer = digit { digit } .  
digit   = "0" | "1" | "2" | "3" | "4" | "5" | '  
while   = "while" "(" expression "  
          ( statement | "{" { statement } "}'
```

= defines. | is *or*. { } is *any number of*, including none. [] is *optional*. () groups. Quotes hold the literal characters. A full stop ends the rule.

So an integer is a digit followed by any number of digits: 0, 42, 007. A while statement is the word, a parenthesised expression, and either one statement or a braced block of them.

The synopsis of week 2 was this notation's cousin, a regular expression: braces, brackets, bars, and nothing else.

EBNF, written in EBNF. Nothing breaks.

```
EBNF      = { production } .
production = non_terminal "=" expression "." .
expression = term { "|" term } .
term       = factor { " " factor } .
factor     = non_terminal | terminal
            | "{" expression "}"
            | "[" expression "]"
            | "(" expression ")" .
non_terminal = variable .
terminal     = "" { character } "" .
```

Every rule on the previous slide is a sentence of this language, these nine included. Check one: integer is a `non_terminal`, an `=`, an expression of one term of two factors, and a full stop.

Nothing strange happens: the text is finite, the reading terminates, the checker is the same stack machine as for any grammar. A text about texts is just another text.

Hold this example. It returns three times, harder each time: a compiler compiling itself, week 8; a program asked about its own halting, week 9; a kernel isolating itself, week 10. The same move on meaning, where the limits live.

C*

7 keywords. 22 symbols. One data type, and pointers to it.

uint64_t

void

sizeof

if

else

while

return

Five statements: assignment, while, if-else, procedure call, return. Arithmetic + - * / %, comparison == != < > <= >=, the * that reads a pointer — hence the star in the name. No other types, no arrays, no Boolean operators.

It is a real subset of C: any C compiler compiles C*. And it is enough to write a compiler for itself, an emulator, and a hypervisor, which is the only argument for its size.

```
// tiny.c
uint64_t c;
uint64_t main() {
    c = 0;
    while (c < 7)
        c = c + 1;
    return c;
}
```

RISC-U

14 instructions. 32 registers. 4 GB of memory. A real subset of real RISC-V.

GROUP	INSTRUCTION	WHAT IT DOES
initialise	lui rd,imm · addi rd,rs1,imm	put a number into a register
memory	ld rd,imm(rs1) · sd rs2,imm(rs1)	load a word from memory, store one
arithmetic	add · sub · mul · divu · remu	rd = rs1 ◦ rs2, unsigned
comparison	sltu rd,rs1,rs2	rd = 1 if rs1 < rs2, else 0
control	beq rs1,rs2,imm · jal rd,imm · jalr rd,imm(rs1)	branch if equal, jump and link, jump to a register
system	ecall	ask the operating system for something

Every instruction is 32 bits. Every one of them also does `pc = pc + 4`, or says where `pc` goes instead. That is the entire machine language, and the machine chapter gives each line its exact meaning.

THREE NOTATIONS, ONE MEANING

What the compiler makes of the **while** loop.

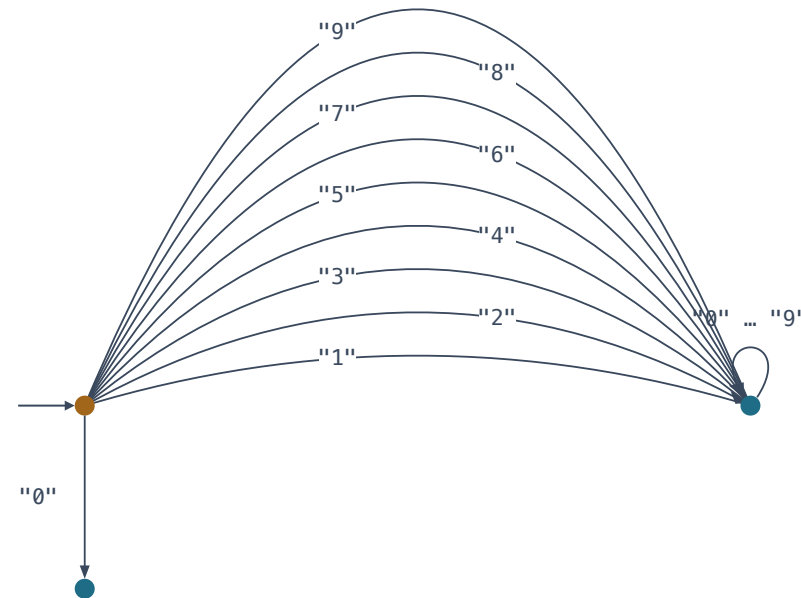
```
$ ./selfie -c tiny.c -S tiny.s
./selfie: 432 bytes generated with 104 instructions and 16 bytes of data
$ cat tiny.s
...
0x158(~6): 0xFF01B283: ld t0,-16(gp)      // c
0x15C(~6): 0x00700313: addi t1,zero,7      // 7
0x160(~6): 0x0062B2B3: sltu t0,t0,t1       // c < 7 ?
0x164(~6): 0x00028C63: beq t0,zero,6[0x17C] // if not, leave the loop
0x168(~7): 0xFF01B283: ld t0,-16(gp)      // c
0x16C(~7): 0x00100313: addi t1,zero,1      // 1
0x170(~7): 0x006282B3: add t0,t0,t1        // c + 1
0x174(~7): 0xFE51B823: sd t0,-16(gp)       // c = ...
0x178(~9): 0xFE1FF06F: jal zero,-8[0x158]  // back to the condition
```

Source. `while (c < 7) c = c + 1; —`
for you.

Assembly. Nine instructions with names
— for the person reading the machine.

Bits. `0xFF01B283` and eight more words
— the only thing the machine sees.
Three notations. One meaning. The last
one is the one that runs.

A finite machine decides a **regular** language.
A stack decides a context-free one.



```

integer      = "0" | non_zero_digit { digit } .
non_zero_digit = "1" | ... | "9" .
digit        = "0" | non_zero_digit .
  
```

integer literals without leading zeros

A finite state machine reads one character at a time and remembers only which circle it is in. That is enough to check integers, identifiers, and the whole synopsis — anything a regular expression describes.

It is not enough for parentheses: to check that every (has its) you must count without bound, and a finite machine cannot. Add a stack and you can: expressions, statements, procedures, all of C*, all of EBNF itself.

Regular inside context-free inside ... — the sizes of notation form a ladder, and week 9 says where the ladder ends.

Everything on this deck is a **finite sequence of symbols**. Hold that thought.

A grammar is a finite text. A program is a finite text. The binary the compiler produced is a finite sequence of bytes. The machine language is a table with fourteen rows.

Finite things from a finite alphabet can be *listed*: all of length one, then two, then three. Next week we do the listing, and it turns out that every program that will ever be written has a number — and that a compiler and an emulator are just arithmetic on those numbers.

THE GAP, SO FAR

Week 2: one prefix, two meanings. Week 3: one byte, five. Week 4: three notations, one meaning, fixed by decree. The decree is a program, and it is written in one of the three notations. That loop is the subject of week 8.

Recommended exercises.

- 1 Read the Notation chapter: C*, RISC-U, EBNF.
- 2 Write the EBNF rule for a C* character literal, and for a string literal. Check against `grammar.md`.
- 3 Derive `c = c + 1;` from the grammar, rule by rule, starting at `statement`.
- 4 Type `tiny.c`, compile it with `-S`, and find the nine instructions of the loop. Change 7 to 700 and see which bits change.
- 5 Write a regular expression for the dates of this course, and an EBNF grammar for nested parentheses. Which of the two needs the stack?

NEXT WEEK

Countability: everything you can write down can be listed, so every program has a number — and a binary is one number, 188,392 bytes long. The compiler compiles itself and gets the same number twice.

Listen, then read.

LISTEN

Mozart · Symphony No. 41 “Jupiter” — Böhm, Vienna Philharmonic, 1979. The finale: a four-note motif and four other themes, combined by the rules of counterpoint in every way the rules allow. A grammar, derived to the end.

READ

Introduction to the Theory of Computation — Sipser: regular and context-free languages done properly, chapters 1 and 2. And, more technical, What can we do about the unnecessary diversity of notation for syntactic definitions? — Wirth, 1977: the one page that gave us EBNF.